SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING
Academic Year: 2023-2024

Program: B Tech/MBA Tech/B.Tech (Intg.)
Stream: (CE/CS/IT/MECH/CIVIL/CSE-CS/AI/EXTC/AIML/AIDS)
Subject: Management Accounting for Engineers
Date: 27 / 04 / 24.
Marks: 100 Marks

Year: I/II/III/IV
Semester:II/IV/VI/VIII
Time: 3 hrs. (10 am to 1 pm)
No. of Pages: 04

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Final Examination/Re-examination (2021-22/2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of the remaining questions, attempt any __4___ questions.
3) In all ___5_ questions to be attempted.
4) All questions carry equal marks.
5) Answer of each new question to be started on a fresh page.
6) Figures in brackets on the right-hand side indicate full marks.
7) Assume Suitable data if necessary.

<table>
<tr><td>Q-1</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>CO- 1; SO- ;
BL- Remember & Understand</td><td>a</td><td>What are the different Branches of Accounting and explain their purposes.</td><td>5</td></tr>
<tr><td>CO-4 ; SO- ;
BL- Evaluate</td><td>b</td><td>A company's total fixed cost is ₹ 80000 for the year 2023. If the product's variable cost and selling price are ₹ 10 and ₹ 15 per unit respectively, find out its Contribution, PV Ratio and BEP (Units & Rs.).</td><td>5</td></tr>
<tr><td>CO-2 ; SO- ;
BL- Evaluate</td><td>c</td><td>The annual demand for an item is 3,200 units. The cost per unit is ₹ 6 and inventory carrying charges are 25% p.a. If the cost of one order is ₹ 150, Calculate:
(a) E.O.Q (Economic Order Quantity)
(b) No. of orders per year
(c) Time between two consecutive orders.</td><td>5</td></tr>
<tr><td>CO-4; SO-;
BL- Evaluate</td><td>d</td><td>From the following information, calculate:
(i) Labour Cost Variance
(ii) Labour Rate Variance
(iii) Labour Efficiency Variance
<table>
<tr><th>Particulars</th><th>Standard</th><th>Actual</th></tr>
<tr><td>Number of hours per unit</td><td>360</td><td>400</td></tr>
<tr><td>Rate per hour (Rs.)</td><td>1.50</td><td>1.40</td></tr>
</table></td><td>5</td></tr>
</table>

<table>
<tr>
<td>Q-2
CO-2; SO-;
BL- Evaluate</td>
<td>a</td>
<td>From the following data extracted from the books of ABC Industries, prepare a Cost Sheet for the year 2023.
<table>
<tr><th>Particulars</th><th>Rs.</th></tr>
<tr><td>Opening stock of raw materials</td><td>50000</td></tr>
<tr><td>Closing stock of raw materials</td><td>80000</td></tr>
<tr><td>Purchases of raw materials</td><td>170000</td></tr>
<tr><td>Carriage inwards</td><td>10000</td></tr>
<tr><td>Direct wages</td><td>150000</td></tr>
<tr><td>Indirect wages</td><td>20000</td></tr>
<tr><td>Other direct charges</td><td>30000</td></tr>
<tr><td>Rent and rates-factory</td><td>10000</td></tr>
<tr><td>Rent and rates – Office</td><td>1000</td></tr>
<tr><td>Indirect materials - Factory</td><td>1000</td></tr>
<tr><td>Depreciation on plant</td><td>3000</td></tr>
<tr><td>Depreciation on office furniture</td><td>200</td></tr>
<tr><td>Office salaries</td><td>5000</td></tr>
<tr><td>Salesman Salary</td><td>4000</td></tr>
<tr><td>Other office expenses</td><td>1800</td></tr>
<tr><td>Other factory expenses</td><td>11400</td></tr>
<tr><td>Managing Director's remuneration</td><td>24000</td></tr>
<tr><td>Other selling expenses</td><td>2000</td></tr>
<tr><td>Travelling expenses</td><td>2200</td></tr>
<tr><td>Carriage outwards</td><td>2000</td></tr>
<tr><td>Sales</td><td>500000</td></tr>
<tr><td>Advance income tax paid</td><td>30000</td></tr>
<tr><td>Advertisement</td><td>4000</td></tr>
<tr><td>Opening stock of finished goods (2000 units)</td><td>80000</td></tr>
<tr><td>Closing stock of finished goods (2000 units)</td><td>------</td></tr>
</table>
Given that, the number of units sold during the year was 10000. The managing director's remuneration is to be allocated as Rs.8000 to the factory, Rs.12000 to the office and Rs.4000 to the selling department.</td>
<td>15</td>
</tr>
<tr>
<td>Q-2
CO-2; SO-;
BL- Remember</td>
<td>b</td>
<td>Explain VED and FSN techniques of Inventory control.</td>
<td>5</td>
</tr>
<tr>
<td>Q-3
CO-5; SO-;
BL- Apply</td>
<td>a</td>
<td>Classic Pen Co. produces and sells two different models of its pens named SMART and EXCELLENT. The selling price of each SMART pen is Rs.90 per pen and its variable cost per unit is Rs.70. Each pen of the EXCELLENT model is sold at a price of Rs.140 and the variable cost per pen is Rs.95. The total fixed costs of the company is Rs. 2,00,000. It is estimated that the volume of sales of the SMART model is approximately 80% of the total sales of the company. Calculate the BEP of each model of pen and the BEP of the company in total (Units & Rs.).</td>
<td>15</td>
</tr>
</table>

<table>
<tr><td>Q-3
CO-4 SO-; BL-Understand</td><td>b</td><td>What is Margin of Safety? Explain with an example.</td><td>5</td></tr>
<tr><td>Q-4
CO- 4; SO-;
BL- Evaluate</td><td>a</td><td>The following information is available in respect of AXE Ltd. and BXE Ltd:
<table><tr><th>Particulars</th><th>AXE Ltd. (Rs)</th><th>BXE Ltd. (Rs)</th></tr><tr><td>Sales</td><td>11,00,000</td><td>14,00,000</td></tr><tr><td>Variable cost</td><td>8,80,000</td><td>10,50,000</td></tr><tr><td>Profit</td><td>1,20,000</td><td>2,00,000</td></tr></table>Calculate:
i. P/V Ratio of both companies
ii. Fixed Cost of both companies
iii. Break Even Point of both companies
iv. Sales to earn a profit of Rs 2,10,000 by each company
v. Margin of Safety of both companies.</td><td>15</td></tr>
<tr><td>Q-4
CO- 2; SO-;
BL- Remember</td><td>b</td><td>Explain any FIVE items that are excluded while preparing a Cost Sheet.</td><td>5</td></tr>
<tr><td>Q-5
CO-3; SO-;
BL-Apply</td><td>a</td><td>Sahara Ltd. gives you the following cost details for the manufacture of product Z at a capacity level of 20,000 units.
<table><tr><th>Particulars</th><th>₹</th><th>Nature of Variability</th></tr><tr><td>Direct Material</td><td>12,00,000</td><td>100% Variable</td></tr><tr><td>Direct Labour</td><td>8,00,000</td><td>100% Variable</td></tr><tr><td>Factory Overheads :</td><td></td><td></td></tr><tr><td>- Lighting and Heating</td><td>4,80,000</td><td>75% Variable</td></tr><tr><td>- Repairs and Maintenance</td><td>3,60,000</td><td>50% Variable</td></tr><tr><td>- Depreciation</td><td>4,80,000</td><td>100% Fixed</td></tr><tr><td>Office Overheads</td><td>2,00,000</td><td>10% Variable</td></tr><tr><td>Selling and Distribution Overheads</td><td>3,00,000</td><td>50% Variable</td></tr><tr><td>Total</td><td>38,20,000</td><td></td></tr></table>Prepare a flexible budget for the production of 15,000 units and 25,000 units, showing total cost and unit cost at each level.</td><td>15</td></tr>
<tr><td>Q-5
CO-3; SO-;
BL- Understand</td><td>b</td><td>What is a Budget and what is its importance? Explain the limitations of Budgeting.</td><td>5</td></tr>
<tr><td>Q-6
CO- 2, SO-;
BL- Apply</td><td></td><td>Rajsekhar Textiles supplies you with the following details from its cost records:
<table><tr><td></td><td>Rs.</td></tr><tr><td>Stock of raw materials on 1 September, 2023</td><td>65,000</td></tr><tr><td>Stock of raw materials on 30 September, 2023</td><td>90,000</td></tr><tr><td>Direct wages</td><td>51,000</td></tr><tr><td>Indirect wages</td><td>3,750</td></tr></table></td><td>12</td></tr>
</table>

	a	Sales	2,00,000	
		Work-in-progress on 1-9-2023	28,000	
		Work-in-progress on 30-9-2023	35,000	
		Purchases of raw materials	75,500	
		Factory rent, rates and power	14,000	
		Depreciation of plant and machinery	3,500	
		Expenses on purchases	2,000	
		Carriage outward	1,000	
		Advertising	5,000	
		Office rent and taxes	2,500	
		Travellers' wages and commission	6,500	
		Stock of finished goods on 1-9-2023	54,000	
		Stock of finished goods on 30-9-2023	31,000	
		Income tax paid	12,000	
		Interest on bank loan	6,000	
		Bad debts	5,000	
		Prepare a Cost Sheet from the above information.		
Q-6 CO-1; SO-; BL- Remember	b	What is Activity-Based Costing? Explain the advantages and disadvantages of Activity Based Costing.		8
Q-7 CO-4; SO-; BL- Evaluate	a	The following details relating to a product are made available to you: **Standard cost per unit :** Material 50 Kg. @ ₹ 40 per Kg. Labour 400 hours @ ₹ 1 per hour. **Actual Cost :** Material 490 Kg. @ ₹ 42 per Kg. Labour 3,960 hours @ ₹ 1.10 per hour Actual production was 10 units. Calculate all Material and Labour variances.		15
Q-7 CO-2; SO-; BL- Evaluate	b	Two components A and B are used as follows: Normal usage: 50 per week each Re-order quantity: A = 300; B = 500 Maximum usage: 75 per week each Minimum usage: 25 per week each Re-order period: A = 4 to 6 weeks; B = 2 to 4 weeks Calculate for each component : a) Re-order level; (b) Minimum stock level; (c) Maximum stock level (d) Average stock level.		5

THE END